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# A new nematode of the genus *Spinitectus* (Nematoda: Cystidicolidae) from the stomach of a hill stream fish *Acanthocobitis* sp.,in Manipur, India

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**Abstract :** A new nematode, *Spinitectus heterospiculi* n. sp., was recovered from the stomach of a fish collected from a hill stream in Manipur, India during the study of helminth parasitic infection in fishes. It differs from its congeners in having different number and arrangement of spines and in the morphology of spicules. It is the first valid record of this genus *Spinitectus* reported from fresh water fish of this region.

**Key words: *Spinitectus*, hill-stream fish, *Acanthocobitis* , Manipur.**

## INTRODUCTION

The nematode genus *Spinitectus* Fourment, (1883)[1] includes a large number of species described from both freshwater and marine fishes, some amphibians, and one mammal (Boomker 1993)[2]. Those species described from the Indian sub-continent are mostly from fishes. Many authors prefer to deal with these parasites according to their zoogeographical regions, i.e. on the basis of their distributions.Sahay, *et al.,*(2005)[3] provided a new key to the species of the genus *Spinitectus*, which includes 27 species of which one species – *Spinitectus ranae* Morishita , (1926)[4], is from an amphibian host and the rest are from fishes. Recent investigations on helminth parasites of freshwater fishes in several hill streams of Manipur, India, revealed a new species of *Spinitectus* in the stomach of a fish *Acanthocobitis zonalternans*(Blyth), which is described in detail.

## MATERIALS AND METHODS

Fishes were collected using a hand net from a hill stream of Gournagar region, Imphal district, Manipur, India during July 2012 and April 2013; the stream joins the Iril river which belongs to the Chindwin-Irrawady systems. The nematodes recovered from the fish were fixed in AFA and transferred in GA for dehydration in a desiccator. Mounting was done in pure dehydrated glycerine. Drawings were made with the aid of camera lucida. All the measurements are in millimetres. Type specimens have been deposited in the Parasitology section museum, Department of Life Sciences, Manipur University.

**Description:**

There are small nematodes with rather thick cuticles bearing rings of spines.Body tapering slightly towards both ends, anterior end truncate.Mouth with indistinct lips. Vestibule rather long; its anterior end dilated to form funnel shaped pro-stoma. First ring is usually situated somewhat posterior to the level of end of vestibule. First ring of spines consisting of 34-36 spines in males and 32-40 spines in females.Rings with large spines extend posterior to region to anterior of glandular oesophagus. Anterior rings more or less regularly spaced; numerous minute spines distributed more or less irregularly in more posterior rings.Spination of cuticle extends posteriorly to some

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distance anterior to coiled posterior body end in male and upto end of tail in female.

**Male** (basedon 5 specimens; measurements of holotype in parentheses):

Length of body 2.90-3.92 (2.92), maximum width 0.14-0.16 (0.15).Length of vestibule 0.07 (0.07), muscular oesophagus 0.18-0.22 (0.18) and glandular oesophagus 0.64-0.74 (0.70). Distance from anterior extremity; first ring of spines at 0.09 (0.09),and nerve ring at 0.11 (0.11). Posterior end of the body ventrally curved. Caudal papillae 10 pairs in number of which 4 pairs are preanal and 6 pairs are postanal. Large (left) spicule slender 0.87-1.08 (0.88) long,with pointed distal end.Small (right) spicule 0.10-0.11 (0.10) long. Length ratio of spicules 8.80:1-9.81:1 (8.80:1). Tail short 0.12-0.15 (0.15).Tail tip bears a mucro.

**Female**(based on 6 gravid specimens):

Length of body3.10-3.77(3.10),maximum width 0.20-0.25(0.20). Length ofvestibule0.07-0.08(0.08),muscular oesophagus 0.19-0.21(0.20),and glandular oesophagus0.67-0.77 (0.67). Distance from anterior extremity, first ring of spines at 0.10-0.11 (0.10),and nerve ring at 0.12-0.15 (0.12). Eggs globular or ellipsoidal, thick shelled without polar filaments. The length of eggs 0.02-0.03, width 0.02. Tail conical and pointed, 0.07-0.08 (0.07).Vulva near anus, at 0.27-0.35

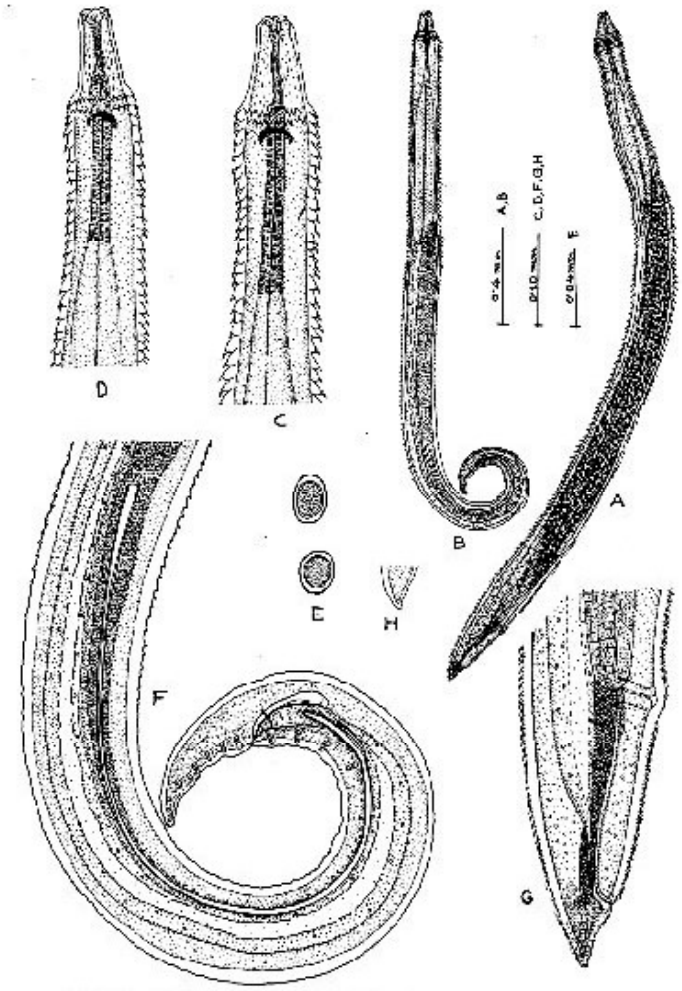



**Fig.1. Camera Lucida drawings of *Spinitectus heterospiculi* n.sp.**

A. Female, whole body, B. Male, whole body, C. Female, anterior end, D. Male anterior end, E. Eggs, F. Male, posterior end showing spicules, G.Female, posterior end showing vulva, H. Male, tail tip of another specimen.

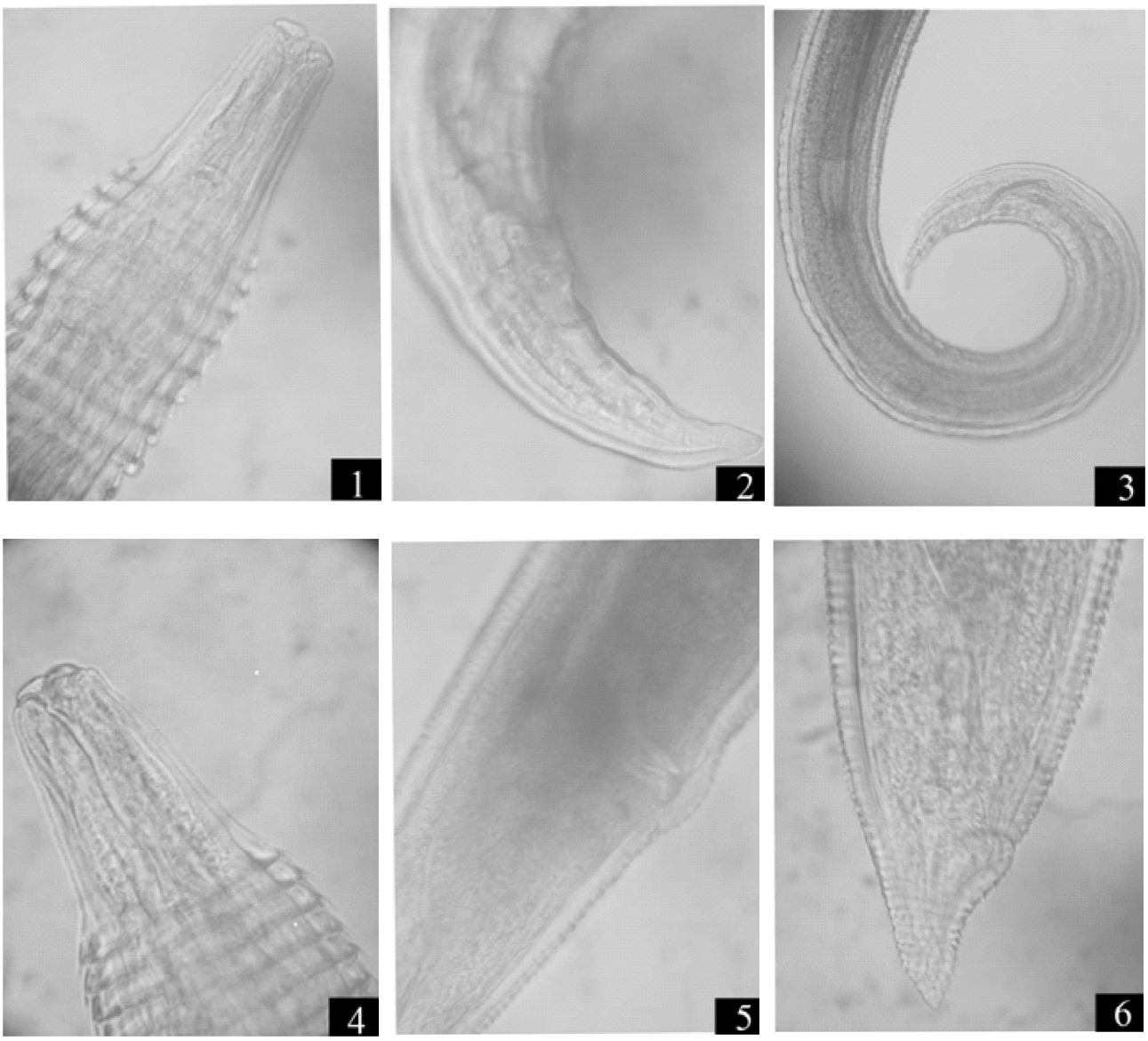



**Plate-1. Photomicrographs of Spinitectus heterospiculi n.sp.**

1. Male, anterior end, 2. Male, posterior end. 3. Male, posterior end showing spicules. 4. Female, anterior end, 5. Female, vulval region, 6. Female, posterior end.

(0.31) from posterior end.Excretory pore just posterior tothe eight- tenth ring of spines (tenth).

## RESULTS AND DISCUSSION

The present specimen belongs to the genus *Spinitectus* as the body is provided with spiny annulations and it meets all the generic characters. The species, when compared with the existing species of the genus *Spinitectus*, resembles none because the spicular ratio is 8.80:1-9.81:1. When compared, in respect of spicular ratio, to its closest species, *S. singhi*(which has a spicular ratio of 8:1), it still differs in many other body characters **(Table 1).** A comparison between the females of the two species is also made where marked differences in the lengths of the body, vestibule, muscular oesophagus, glandular oesophagus are found. Positions of the nerve rings are

different (**Table 2).** It closely resembles *S. bengalensis* with regard to body dimensions. However, their spicular ratios are markedly different. For the present specimen, the ratio is 9:1 while that of *S. bengalensis* is 5:1 **(Table1).**Females of the two species show differences **(Table 2).** The present specimen falls in the category where eggs are without polar filaments, spicules are unequal and dissimilar. *S. yorki* falls under the same category, but it has 20 pairs of caudal papillae. The other species under the above the category where the number of caudal papillae is less than 20 pairs are- *S. bancroftei* ( the spicular ratio 3:1), *S. singhi* (8:1), *S. corti* (5:1), *S. pandharinathi* (4:1), *S. komiyai* (2:1). The spicular ratio, nevertheless, in the present nematode is 8.8:1. Therefore, the present specimen is considered to be new and hence, a name has been given as *Spinitectus heterospiculi* sp. nov. to accommodate the new species. Sahay (1970)[5] has considered the following characters of importance while considering and dealing with the nematodes of the genus *Spinitectus* for proposition to be a new one, viz., the spicules, caudal alae, male tail and the number of caudal papillae, muscular and glandular oesophagus, and the position of vulva.

**Table 1: A comparison between the males of *S. heterospiculi*n.sp.,*S. singhi* Ali, 1956 and *S. bengalensis*Chakravarty, Sain, and Majumdar, 1961**

| Name of the organs | *S. heterospiculi*n.sp. | *S. singhi* | *S. bengalensis* |
|---|---|---|---|
| Body | 2.90-3.24x 0.14-0.16 | 6.97 x 0.08 | 2.85-3.36 x 0.15-0.16 |
| No. of spiny rings | **Not lesser than 230** | **143** | **10** |
| No. of spines in 1st ring | 34 – 36 | 24 | **22** |
| Vestibule | 0.07x 0.01 | 0.05 | - |
| Muscular oesophagus | 0.18-0.22x 0.02-0.03 | 0.39 | 0.19-0.20 x 0.02-0.03 |
| Glandular oesophagus | 0.64-0.74x 0.07-0.08 | 2.21 | 0.40-0.55 x 0.06-0.07 |
| Nerve ring | 0.11 | 0.12 | 0.19-0.20 |
| Caudal papillae | 4+0+6 | 4+0+5 | 4+0+7 |
| LS:RS | **9:1** | **8:1** | **5:1** |
| Tail | 0.12-0.15 | 0.11 | 0.09-0.10 |

Host-*Acanthocobitis zonalternans*, Locality – Gournagar (Manipur)
Site of infection-Stomach, Number of fish examined- 12,
Number of fish infected- 01, Number of parasites recovered -16
(7 males and 9 females)

**Table No.2: A comparison between the females of *S. heterospiculi* n. sp.,*S. singhi* Ali, 1956 and *S. bengalensis*Chakravarty, Sain, and Majumdar, 1961**

| Name of the organs | *S. heterospiculi* n.sp. | *S. singhi* | *S. bengalensis* |
|---|---|---|---|
| Body | 3.10-3.77 x 0.20-0.25 | 3.94-4.47 x0.05-0.07 | 2.4-3.56 x 0.18-0.30 |
| No. of spiny rows | **Not lesser than 340** | **83** | **12** |
| No. of spines in 1st row | **32-40** | **26** | **22** |
| Vestibule | 0.07-0.08x0.01 | 0.03-0.04 | - |
| Muscular oesophagus | 0.19-0.21 x 0.02-0.03 | 0.37-0.47 | 0.14-0.15 x 0.02-0.03 |
| Glandular oesophagus | 0.67-0.77 x 0.08-0.09 | 1.04-1.11 | 0.42-0.56 x 0.04-0.07 |
| Nerve ring | 0.12-0.15 | 0.09 | 0.15-0.20 |
| Vulva | 0.27-0.35 | 0.25-0.32 | 0.14-0.27 |
| Tail | 0.07-0.08 | 0.06-0.07 | 0.02-0.07 |

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